Special issue: Urban and ageing population

Urban Studies

Urban Studies
1–22

DOI: 10.1177/00420980261460329
journals.sagepub.com/home/usj
Sage

A caring smart city: A bottom-up approach with older adults in Santiago de Chile

Francisco Vergara-Perucich[1]

Abstract

Age-friendly and smart-city agendas have developed in parallel: the former foregrounds older adults' wellbeing but lacks operationalisation, with indicators under-capturing digital inclusion and pedestrian safety; the latter deploys data platforms treating older people as passive subjects within privately governed systems. Although care is increasingly recognised as a principle of urban governance, few studies examine how co-designing civic technologies with older adults can generate context-sensitive spatial metrics for municipal decision-making. This article asks: how can digital technologies be repurposed as infrastructures of care to strengthen older people's right to the city in Global South contexts? Drawing on the Ciudad Mayor project (2022–2024) in three municipalities of Santiago de Chile, a sequential mixed-methods design is employed: participatory collective mapping with older residents; iterative co-design and usability testing of a mobile app; and geospatial analysis producing '"hyper-cartographies"' — maps combining citizen-generated reports with objective indicators. Participatory mapping revealed micro-barriers — such as uneven pavements and poor lighting — that escape standardised indicators but condition everyday mobility. The co-designed app functioned as a traceable channel linking older people's situated knowledge to neighbourhood budgeting. Older participants demonstrated strong capacity to engage with accessible digital tools. Conceptually, infrastructures of care are positioned as a bridge between age-friendly and smart-city debates. Methodologically, a replicable protocol integrating co-design with Geographic Information Systems (GIS) is proposed to produce context-sensitive metrics, reframing the smart city as a democratic, age-affirming project rather than a techno-managerial fix.

Keywords

age-friendly cities, care ethics, Global South, older adults, participatory GIS, smart city

[1]Universidad de Las Américas, Santiago, Chile

Corresponding author:
Francisco Vergara-Perucich, Centro Producción del Espacio, Universidad de Las Américas, Santiago 7500972, Chile.
Email: jvergara@udla.cl

摘要

适老化城市议程与智慧城市议程同步推进：前者将老年人福祉置于核心，但缺少落地实施机制，现有指标难以充分反映数字包容性与步行安全；后者部署各类数据平台，在私营治理体系中将老年人视作被动客体。尽管照护日益被视作城市治理的一项原则，但鲜有研究探讨与老年人协同设计公共技术如何生成在地适配的空间指标，服务市政决策。本文提出核心问题：在全球南方背景下，如何将数字技术转型打造为照护基础设施，以加强老年人的城市权？本文依托智利圣地亚哥三座市镇 2022 至 2024 年实施的《老年城市》(Ciudad Mayor) 项目，采用序贯混合研究设计：面向老年居民开展参与式集体制图，迭代完成移动端应用的协同设计与可用性测试，并借助地理空间分析生成“超制图”，即将公民生成报告与客观指标相结合的地图。参与式制图揭示了不平整的路面以及照明不足等微观障碍，这些障碍未能被标准化指标覆盖到，但却制约着日常出行。协同设计的应用程序充当可追溯的渠道，将老年人的在地经验知识与街区财政预算编制联系起来。老年受访者展现出较强的能力，可顺畅使用无障碍数字工具。从概念层面而言，照护基础设施被定位为衔接适老化城市与智慧城市相关理论讨论的桥梁。从方法论层面而言，本文提出了将协同设计与地理信息系统 (GIS) 相结合的可复现研究方案，用以生成在地适配的指标，将智慧城市重新定义为民主、积极适老的项目，而不是技术管控型治理手段。

关键词

适老化城市、照护伦理、全球南方、老年人、参与式地理信息系统、智慧城市

Received: 28 January 2026; accepted: 27 May 2026

Introduction

Over the past two decades, the age-friendly turn in urban design has brought older adults to the centre of territorial planning. Yet the operationalisation of that turn remains contested: indicator frameworks under-capture digital inclusion, pedestrian safety and food environments, and they rarely engage digital infrastructure as constitutive of urban life rather than as a peripheral amenity (Kim et al., 2020; Rugel et al., 2022; Pinheiro et al., 2018). Simultaneously, the smart-city paradigm — understood here as the techno-managerial model of urban governance that deploys sensors, platforms and data analytics to optimise urban flows, typically under public–private partnerships oriented to economic competitiveness (Kolotouchkina et al., 2024) — has enrolled older adults as passive data nodes within privately governed systems, treating them as objects of monitoring rather than as epistemic subjects capable of co-producing urban knowledge.

This article reframes the smart city through an ethics of care, taken as an organising principle for the production of space and for urban governance, with specific attention to contexts in the Global South. The conceptual premise is Lefebvrian by asserting that the city cannot be merely seen as a container of functions. Instead, it is a complex network of social relations, shaped by political and technocratic tensions and capitalist modes of production (Lefebvre, 1991, 2004). The urban exceeds the city once production is territorialised and everyday life is subordinated to the pursuit of economic performance, reconfiguring the socio-spatial division of labour (Monte-Mór, 2005). Within these dynamics, older adults — especially retirees — are frequently positioned at the margins insofar as they sit outside direct circuits of economic production (Phillipson et al., 1999; Osorio Parraguez et al., 2011). It is therefore pertinent to read the smart-city paradigm from the standpoint of those whom it risks excluding.

Research on ageing and the city consistently associates residential quality, neighbourhood conditions, and perceived social security with physical and mental health in old age (Liu et al., 2017). Community ties and local proximity buffer social and biographical change in later life (Phillipson et al., 1999), and recent proximity-based agendas, such as the 15-minute city, revisit how amenities and services are ordered to reduce everyday frictions (Moreno, 2020; Ulloa-León et al, 2023). International evidence sharpens the analytical frame. In Singapore, expansion of the Circle Line reconfigured older people's activity and travel patterns through combined mobility and amenity effects (Zhu and Diao, 2025); in the United States, aging in place in gentrifying neighbourhoods is associated with adverse self-rated and mental health outcomes, particularly among economically vulnerable older adults (Smith et al., 2018), and apparent advantages for older renters under rent control appear to be mediated by tenure stability rather than age per se (Bian et al., 2026); in China, both individual and neighbourhood educational context are associated with lower depressive symptoms in old age (Li and Zhao, 2021). These dimensions — mobility reconfiguration, tenure stability and neighbourhood socio-educational context — inform the analytical variables we apply to the Santiago case.

A normative lens emerges at the confluence of care, feminist urbanism and the right to the city. Cities have been historically and materially produced as masculinised artefacts; a feminist approach recentres embodied experiences of women and marginalised groups at the core of urban projects (Kern, 2020; Pojani, 2021), whilst remaining attentive to the risks of instrumentalising the feminine as place-marketing that may fuel displacement (van den Berg, 2012). In rights terms, proposals to extend the right to the city with a gendered dimension emphasise belonging, safety and everyday empowerment (Fenster, 2005). These arguments align with compassionate- and healthy-city agendas, where investment in health, social services and active citizenship underpins inclusion and equity (Green et al., 2015; Flores et al., 2018), and with discussions of a municipalism of care that seeks to institutionalise urban commons and neighbourhood solidarities without reducing them to rhetoric (Davis, 2022; Kussy et al., 2022).

Despite this growing convergence, few studies have examined how participatory co-design of civic digital tools with older adults can generate context-sensitive spatial metrics that feed directly into municipal decision-making. Smart-city scholarship cautions that unreflective digitisation risks deepening social and digital inequalities (Kolotouchkina et al., 2024); care scholarship identifies shadow infrastructures that sustain everyday life in post-welfare cities (Power et al., 2022); co-design reviews highlight the value of workshops and multi-actor participation but note gaps in implementation traceability and policy impact (Sanz et al., 2021); and usability syntheses for older users emphasise simplification of controls but rarely connect these practices to urban metrics of mobility and safety (Gomez-Hernandez et al., 2023). The alignment between technical usability and the lived requirements of safe urban mobility remains under-theorised and under-measured.

This article addresses that gap. Drawing on the Ciudad Mayor project (2022–2024), implemented in the working- and middle-class municipalities of Renca, Independencia and Quinta Normal in Santiago de Chile and funded by the Santiago Regional

Government's innovation programme, we advance three research questions:

RQ1: What micro-scale spatial frictions condition everyday mobility and social participation of older adults in working- and middle-class municipalities of Santiago?

RQ2: How can a co-designed civic digital tool translate older adults' situated experiential knowledge into actionable inputs for municipal planning and neighbourhood budgeting?

RQ3: Under what institutional and infrastructural conditions can a digitally mediated commons support a care-oriented smart city in the Global South?

The article offers three contributions. Conceptually, it brings infrastructures of care into conversation with age-friendly and smart-city debates, positioning older adults as epistemic subjects and co-producers of space. Methodologically, it proposes a replicable protocol that braids qualitative co-design (workshops, collective mapping, smartphone-application user testing) with quantitative geospatial analysis to produce what we term *hyper-cartographies:* multilayer maps that combine citizen-generated georeferenced reports, qualitative coding of situated experience and objective spatial indicators (tree cover, pavement condition, lighting, service proximity) into composite visualisations for municipal decision-making. Politically, advancing the co-designed application to Technology Readiness Level 6 (TRL-6) opened a pilot channel through which older people's reports could inform neighbourhood planning discussions; the process also unsettled stereotypes of digital aversion among older adults, who, accelerated by the pandemic, engaged with notable appetite for formative collaboration.

The article proceeds as follows. The next section reviews the literature at the intersection of age-friendly metrics, digital inclusion and care-centred urbanism. We then situate the study within the Chilean urban context before presenting the sequential mixed-methods design. Results are organised around participatory mapping, the digital module and system architecture, hyper-cartographies and institutional adoption. The discussion engages three themes: the micro-scale of exclusion, the digital divide as infrastructure, and the institutional conditions for governing care. Final reflections draw out implications for policy and comparative research.

Literature review

The intersection of ageing, technology and care in cities has shifted away from technocratic smart-city framings towards approaches that foreground interdependence, agency and spatial justice. This section reviews three strands of that shift: the limitations of current age-friendly indicator systems; the state of digital inclusion, co-design and health technology for older populations; and the emerging articulation of territorial accessibility with care ethics and urban commons.

Age-friendly metrics: Gaps and limits

A persistent disconnect is visible between the WHO's age-friendly agenda and the instruments municipalities actually deploy. Evidence from the AARP (formerly the American Association of Retired Persons) survey identifies 57 indicators and a three-factor structure centred on the built environment, transport and the social environment, with positive associations between these domains and older people's physical health; yet digital dimensions, 15-minute accessibility, pedestrian safety and food environments are rarely integrated explicitly (Kim et al., 2020). Cross-national analyses across 20 countries recommend adapting indicators to degrees of urbanity and income levels, cautioning against one-size-fits-all approaches and domain scores that mask complex relationships among variables (Rugel et al., 2022). Recent validation of age-friendly measurement instruments across age cohorts confirms that financial dimensions significantly shape perceived age-friendliness, especially in domains of social participation and community health services, and that age-friendliness can and should be measured intergenerationally rather than as an exclusively older-adult concern (Marston et al., 2026). Expert hierarchisations tend to privilege active ageing concerns — physical activity, social engagement, productive participation — while leaving inclusionary digital access, pedestrian safety, and nutrition outside the operational core of assessment (Pinheiro et al., 2018). A broader critical stocktaking of the age-friendly cities movement identifies persistent questions about built-environment integration, noting that the intersection with smart-city agendas has been a missed opportunity and that a global overhaul of the age-friendly model is overdue (van Hoof et al., 2021).

These limitations extend beyond mere technical constraints. When viewed through a Lefebvrian lens, indicator systems actively contribute to the social construction of abstract space. They render cities comprehensible to planners and investors by categorising lived experiences, thereby perpetuating the socio-spatial division of labour (Lefebvre, 1991; Monte-Mór, 2005). Consequently, when older adults' embodied frictions — such as a broken pavement, a perceived too-short crossing, or a poorly lit stretch that evokes gendered fear — are absent from the metric, the instrument itself becomes a tool of exclusion. Compounding this structural critique, recent investigative analysis reveals that the eight WHO age-friendly topics[1] were never formally validated prior to 2020 and were rooted in North American elder-friendly literature without transparency about that origin, producing downstream failures in measurement instrument development and robust critiques of blind spots such as financial dimensions and safety (van Hoof and Marston, 2025). Current age-friendly frameworks thus risk erasing local complexity behind technocratic simplifications, a concern that extends to digitally mediated smart-city dashboards unless their indicator architectures are grounded in situated knowledge.

Digital inclusion, co-design, and health technology

At the interface of digital inclusion, participatory design, and public health, progress is uneven. A systematic review on digital competences and access to justice indicates that older adults' technological literacy conditions their agency in public services and their capacity to exercise rights, pointing to the need for age-sensitive state policies for inclusion (Masías Benavides Román and Chipana Fernández, 2021). In health promotion and prevention, a broad scoping review of digital technologies notes potential benefits for mobility, mental health and nutrition, but calls for more diverse samples, outcome evaluations that move beyond feasibility pilots and human supports to scaffold adoption (De Santis et al., 2023). Earlier foundational work on ambient intelligence technologies for ageing-in-place found that older adults' primary motivation was safety and security — particularly fall detection — and that these systems must integrate with existing care provision rather than substitute it, a finding that anticipates current debates on co-design and institutional uptake (van Hoof et al., 2011). Reviews of co-design in people-centred digital solutions highlight the value of workshops, interviews and multi-actor participation to align needs with prototypes, while noting persistent gaps in participant numbers, implementation traceability and policy impact evaluation (Sanz et al., 2021). Usability syntheses for older users emphasise simplification and increased size and spacing of interactive controls, alongside guidance on navigation, cognitive load and training; however, these practices are rarely connected to urban metrics of mobility and safety (Gomez-Hernandez et al., 2023).

Smart-city scholarship reinforces the concern. Unreflective digitisation risks deepening social and digital inequalities in the absence of comprehensive inclusion frameworks and governance logics that extend beyond operational efficiency (Kolotouchkina et al., 2024). Systems-oriented proposals offer a partial corrective: the social–ecological–technological systems (SETS) perspective argues that interactions across social, ecological and technological domains can reorient smart agendas toward sustainability and equity, providing tools to measure change and mobilise diverse actors in urban transformation (Branny et al., 2022). Yet the practical integration of these perspectives into participatory civic technologies designed with — rather than for — older adults remains limited.

Territorial accessibility, care, and urban commons

Territorial accessibility and everyday mobility sit at the core of ageing in the city. Analyses of 15-minute proximity in Santiago report broad potential coverage for older residents but identify gaps in public policy, health provision and public-space design (Ulloa-León et al., 2023). Neighbourhood-scale qualitative work documents intermodal strategies, preference for local destinations and the delegation of tasks, underscoring how older people's mobility is shaped by relational networks and proximate offerings (Vecchio et al., 2020). A systematic review of public-transport accessibility for older adults finds substantial methodological variability and calls for engagement with critical gerontology and age-friendly frameworks to standardise assessments of mobility and social participation (Ravensbergen et al., 2022). In Santiago specifically, pedestrian morphology, vegetation, façades, and cleanliness influence willingness to walk, while degraded pavements and insufficient lighting are associated with fear and accidents among older residents (Herrmann-Lunecke et al., 2022).

Care scholarship sharpens the analytical frame. Recent work identifies shadow infrastructures — informal and often precarious networks of mutual aid, domestic labour and neighbourhood solidarity — that sustain everyday life in post-welfare cities (Power et al., 2022). The question is whether and how public institutions can formalise such practices without co-opting or hollowing them out. Experiments in a municipalism of care examine how local administrations in cities such as Barcelona have sought to institutionalise urban commons and neighbourhood solidarities through participatory budgets, care cooperatives, and civic technology platforms, with tensions between ambition and fiscal-administrative constraint (Kussy et al., 2022). Ethical perspectives on urban management argue that care should operate as a design and governance principle rather than as a narrow sectoral remit, thereby widening the circle of actors involved and accountable (Davis, 2022). The Healthy Cities tradition converges on this point, linking investment in health, social services, active citizenship and health literacy to greater inclusion and equity (Green et al., 2015).

More broadly, residential and neighbourhood conditions, coupled with safe social environments, are associated with physical and mental health outcomes in old age (Liu et al., 2017). Community life buffers social change in later life, yet the quality of that buffering depends on the material fabric of the neighbourhood and on the institutional arrangements that sustain it (Phillipson et al., 1999; Woolrych et al., 2021). Age-friendly community agendas have sought to formalise this buffering through co-design, wider participation and multi-sectoral collaboration, yet their aspirational framework requires a systematic overhaul to address growing inequality, digital transformation and the persistent gap between WHO indicators and the instruments municipalities actually deploy (Buffel and Phillipson, 2018). The pandemic exposed and amplified these mechanisms, restricting rights and public-space use for older people through protective enclosure that narrowed everyday sociability (Rojas et al., 2021).

The Chilean context

In 2019, the country witnessed a nationwide social upheaval, initially ignited by protests against public transport fares. However, the underlying cause was profound concerns about inequality and a critical examination of the prevailing neoliberal model (Azócar González, 2023; Cox et al., 2024). This period of unrest revealed a highly fragmented social landscape, marked by a stark ideological divide among elites. These elites often responded to social demands with resistance, fear-mongering, and the caricaturing of others, rather than engaging in transformative dialogue (Gayo and Méndez, 2022; Pelfini et al., 2023).

Against this backdrop of acute social conflict, the COVID-19 pandemic introduced unprecedented challenges. Public health promotion policies implemented in Chile during the crisis primarily focused on economic measures that carried significant health and broader social implications (Ramírez-Pereira et al., 2020). For older adults residing in Santiago, the ability to safely navigate the city and access local destinations was already a critical determinant of quality of life and social participation (Vecchio et al., 2020) and the pandemic severely disrupted this daily reality. It significantly decreased physical accessibility to health services and forced a rapid, unguided transition to online platforms, which, combined with forced social isolation, had a profound impact on the mental health and everyday behaviour of the ageing population (von Humboldt et al., 2022).

As essential services digitised rapidly, a severe digital divide became glaringly apparent. Socio-economic and demographic factors present significant barriers for older adults attempting to access digital services, including e-Justice and healthcare platforms (Loyola, 2019). Consequently, there is an urgent need for targeted public policies and digital literacy programmes to bridge this generational gap (Riveros et al., 2020). Qualitative evidence from Chilean initiatives demonstrates that when older adults are supported through practical, 'learning by doing' strategies, digital inclusion can significantly enhance their autonomy, self-efficacy, and civic empowerment (Valenzuela Urra et al., 2022). Therefore, integrating digital inclusion with urban mobility improvements presents a vital pathway to fostering a caring city, equipping older adults with the tools to navigate this complex socio-political landscape.

Synthesis

Across these intersecting strands, the literature signals a critical shift from a technocratic smart city to a care- and justice-centred urbanism that positions older adults not as passive beneficiaries but as epistemic agents. Age-friendly metrics, as currently configured, are partial and context-insensitive; digital divides constrain rights and co-design literatures have yet to establish robust impact pathways at the municipal scale; and territorial accessibility, whilst promising in proximity terms, falters where the supporting infrastructure of care — pavement quality, lighting, shade, resting points, and human support for digital adoption — is absent or unevenly distributed. As the Chilean context demonstrates, these systemic shortcomings are amplified by structural inequalities, post-pandemic isolation and a stark digital divide, making the need for situated, bottom-up interventions particularly urgent. The emerging agenda thus demands a governance of care that aligns civic technology, community infrastructure and urban design to support autonomy and the right to the city in later life. To address this gap, the following section outlines the sequential mixed-methods design deployed to operationalise this caring smart-city agenda in Santiago de Chile.

Methods

The research employed a sequential explanatory mixed-methods design. Participatory qualitative inquiry informed the operationalisation of variables, which were then examined using geospatial cartography. Theoretically, the study is rooted in feminist urbanism and spatial justice. It positions older adults as epistemic subjects, rather than passive service recipients, and views knowledge as co-produced through the interaction of participants, facilitators, and technology. In practice, this work stems from a publicly funded R&D project focused on municipal technology transfer. The project aims to develop a platform, mobile app, and dashboard with a strong emphasis on usability and institutional adoption.

Study sites and participant sampling. Fieldwork took place in three municipalities of Santiago — Renca, Independencia and Quinta Normal — characterised by working- and middle-class profiles and by urban heterogeneities relevant to mobility and access for older residents. Participation was formalised through municipal agreements and designated technical teams for co-design and validation, under the Santiago Regional Government's Regional Competitiveness Innovation Fund (FIC-R), Project "Ciudad Mayor", implemented between 2022 and 2024.

Sampling combined focus groups, collective mapping workshops with older adults, co-design sessions with municipal staff, and usability tests. The participant distribution across the primary study sites was structured as follows:

- *Total participants:* 30 older adults (ten per municipality).
- *Demographics:* Purposive variation by gender, age bands within the 60 + cohort, and specific neighbourhood trajectories.
- *Engagement cycles:* Multiple iterative rounds including initial qualitative workshops, interface co-design sessions, and subsequent usability testing with technical personnel and a subgroup of older users.

Sequential design and data collection. The research followed a linear flow: qualitative mapping → app co-design → geospatial integration → institutional piloting. Workshops gathered situated experiences where participants traced everyday routes, points of pleasure and fear, micro-scale obstacles (potholes, unsafe crossings, heat islands) and urban supports (shade, seating, toilets). These materials were used to construct a socio-spatial lexicon complementing WHO age-friendliness domains.

The qualitative corpus directly informed the specification of app functions and the municipal dashboard. Incremental prototypes of the mobile app and an interactive analytics dashboard were implemented, with iterations guided by usability sessions. Tests assessed flow comprehension, typographic legibility, action hierarchy and system feedback, as well as friction on key tasks (reporting pavement/lighting issues, locating services).

Hyper-cartographies and spatial analytics. We use the term *hyper-cartographies* to designate multi-layer maps that combine citizen-generated georeferenced reports, qualitative coding of situated experience, and objective spatial indicators (tree cover, pavement condition, lighting, service proximity) into composite visualisations for municipal decision-making. Unlike conventional thematic cartography, hyper-cartographies foreground experiential data co-produced with residents, enabling the overlay of perceptions of safety, walkability and access with material infrastructure conditions.

To process this data, triangulation proceeded in two stages. First, thematic analysis of workshops and focus groups coded emergent dimensions (everyday mobility, affects and fears, micro-obstacles, care supports). Qualitative data were processed using thematic coding software to establish categories and ensure inter-coder reliability, maintaining traceability between inductive categories and normative WHO frames. Second, in parallel with the geographic database assembly, geospatial modelling and spatial analysis routines were implemented using R and Python. These scripts calculated pedestrian accessibility (distance/time thresholds, slope gradients), road safety and ambient conditions, ultimately translating the qualitative lexicon into context-sensitive metrics.

To enhance robustness, we convened an expert panel (Delphi-type procedure) comprising specialists in urban planning, ageing, and municipal management to appraise the metrics and propose criteria for prioritising neighbourhood investment.

The Providencia institutional pilot. Following the participatory and co-design phases in Renca, Independencia and Quinta Normal, we conducted a municipal-scale pilot in Providencia to test the platform and dashboard under live management conditions (budget hearings, responses to residents' requests). Providencia was selected for this stage because its municipal administration had expressed institutional willingness and possessed the technical capacity to integrate the dashboard into existing workflows — conditions that were not yet in place in the three originating municipalities at the time of piloting.

The Providencia–Nueva Providencia corridor (Miguel Claro Street to Tobalaba Avenue) offered a high-density, mixed-use environment that exercised the full range of platform functions. We note that Providencia's higher-income profile means that piloting results reflect institutional and technical feasibility rather than the socio-spatial conditions documented in the participatory phase. Findings from the three originating municipalities and from the Providencia pilot are therefore presented and interpreted separately.

Ethics and data governance. The study was approved by the ethics committee of Universidad de Las Américas (UDLA) under the "Ciudad Mayor" (FIC-R) project framework. All participants provided written informed consent prior to participation in workshops, co-design sessions and usability tests. Consent forms were adapted for readability, using larger font sizes and plain language, and were read aloud to participants who requested assistance. Specific safeguards were implemented for participants with reduced functional autonomy, including the option for a trusted companion to be present during sessions.

Georeferenced data captured through the application were stored in anonymised form: user accounts are not linked to national identity numbers, and report coordinates are associated with the reported location rather than the user's residential address. The analytics platform operates under specific terms of use, and access is restricted based on role-based profiles, such as citizen, municipal, or administrator/researcher. Photographic records of workshops and field testing were obtained with explicit consent, and identifying facial features were blurred in publication materials to comply with double-blind review protocols and protect participant privacy.

Limitations

Several limitations should be acknowledged. First, the sample was purposive and small, comprising approximately ten participants per municipality in workshops, which restricts generalisability. Self-selection is likely, as participants who attended were probably more socially active and digitally inclined than the broader older population. Second, the study was conducted in a single metropolitan area, making it difficult to assess the transferability of the findings to cities with different land regimes, institutional capacities, or demographic compositions. Third, the application prototype reached TRL-6 but has not been evaluated for long-term adoption, sustained municipal use, or measurable impact on budgetary outcomes. Fourth, the institutional pilot was conducted in Providencia, a municipality with a distinct socio-economic profile from the three study sites, which limits direct comparability of adoption conditions. Fifth, the Android-only release excluded older adults using iOS devices, a constraint driven by cost and prevalence in the study communities but one that narrows the scope of the study. Finally, the absence of a control group or counterfactual design prevents us from attributing observed changes in municipal responsiveness solely to the platform.

Results

Participatory mapping

Addressing RQ1, participatory mapping served as the starting point for translating older adults' situated experiences of micro-scale spatial frictions into operational inputs. Workshops were held across Renca, Independencia and Quinta Normal using printed neighbourhood maps (1:7500–1:10,000) and a deliberately simple visual language of coloured stickers to rate routes and specific points against agreed categories (perceived safety, pavement continuity, lighting, shade, crossings and service access).

The protocol moved from collective activation — grounding labels in everyday situations — to small-group mapping of pleasure, fear and micro-barriers, concluding with a plenary synthesis. This sequence enabled a shift from dispersed narratives to verifiable spatial configurations. A consistent repertoire of infrastructural and affective frictions emerged: solitary and poorly lit stretches, degraded pavements, insufficient signal times, and deficits of shade.

To capture the density and nature of these frictions across the diverse municipal contexts, Table 1 summarises the workshop outputs.

Participant accounts added a layer of nuance that spatial tags alone could not capture, showing how gaps in infrastructure directly shape where older adults can and cannot go, and on what terms. One participant in Independencia put it plainly:

> The transport system has not adapted to the changes in the municipality; for example, after the metro arrived, the colectivos kept running the same routes, and the buses too, with no new stops and no changes since the metro came. (Male participant, Independencia, May 2023)

The gendered texture of safety came through just as clearly in Quinta Normal: 'I wouldn't

Table 1. Workshop outputs.

Municipality	Workshops (*n*)	Total participants	Key micro-barriers identified	Most tagged categories
Renca	2	∼20 (≈10 per session; 5 women, 5 men, ⩾ 1 migrant per focus group)	• Poorly lit and solitary stretches conditioning fear-driven route avoidance • Degraded pavements • Bus stop placement and design incompatible with older users • Garbage accumulation near key pedestrian routes • Plazas and parks perceived as unsafe due to alcohol consumption and informal gatherings • Lack of shade on main corridors	• Perceived safety/ lighting • Pavement continuity • Bus stop accessibility • Public space safety
Independencia	2	∼20 (≈10 per session; 5 women, 5 men, ⩾ 1 migrant per focus group)	• Public transport routes not adapted to metro network arrival (colectivos/ buses maintaining pre-metro corridors without new stops) • Poorly lit streets and avenues • Degraded pavements • Bus boarding steps incompatible with older adults' mobility • Insufficient pedestrian crossings	• Transport adaptation • Lighting • Pavement condition • Bus boarding accessibility • Crossing safety

(continued)

Table 1. (continued)

Municipality	Workshops (*n*)	Total participants	Key micro-barriers identified	Most tagged categories
Quinta normal	2 (1st in-person; 2nd remote)	~20 (≈10 per session; 2nd session conducted telematically)	• Plazas and parks perceived as sites of drug trafficking and alcohol consumption (Plaza Besa cited most frequently) • Narrow pavements occupied by cyclists creating pedestrian conflicts • Bus stops in conflict with cycle lane (ciclovía) infrastructure • Poor lighting on main avenues (Av. Carrascal; Costanera Sur) • Solitary and dark stretches triggering avoidance behaviour • Inaccessible bus boarding • Lack of pedestrian ramps	• Safety/perceived plaza use • Cycling–pedestrian conflicts • Bus stop design • Lighting • Pavement continuity

Note: The second participatory session in Quinta Normal (usability testing, May 2023) was conducted remotely owing to the municipality's concern about exhausting the participatory capital of its older residents' organisations. All workshops used printed neighbourhood maps (1:7500–1:10,000), coloured stickers and post-its as the primary elicitation tools. Participants additionally completed smartphone usability tests of the Ciudad Mayor application; across Renca and Independencia (*n* = 20), 92% identified the app's main functions quickly, 83% navigated between sections easily, and 75% considered button size adequate, while 42% reported fluency issues attributed to connectivity infrastructure rather than application design.

walk along there [Av. Carrascal] because it has no lighting. Not at any hour' — and later in the same session — 'That square is dangerous — there's a lot of drug trafficking and they drink a lot. I avoid it' (Female participant, Quinta Normal, focus group 1, 2022).

These perceptions, gathered in the early workshop rounds, shaped the app's specification in concrete ways: they informed the taxonomy of reportable incidents and guided decisions about which options to prioritise in the interface hierarchy. Workshop deliberation was structured to keep any single voice from dominating: spokesperson roles rotated, turn-taking was monitored, and dissenting views — such as the divergent daytime and night-time readings of the same square — were recorded rather than resolved. Facilitators held back from translating accounts into technical categories too quickly; meanings were validated in the group's own terms before any correspondence with

Figure 1. Photograph of the app being tested in the field with residents of Independencia.
Source: Author.

institutional WHO age-friendliness domains (mobility, outdoor spaces, social participation, services) was proposed. This mattered especially in discussions of fear of crime and street harassment, where gendered distinctions that emerged from the accounts shaped both subsequent lines of enquiry and specific design decisions.

From each table session, three tangible outputs emerged: a physical map with markers prepared for later georeferencing (through digitisation and anchoring to the base cartography); brief minutes capturing agreements and priorities; and a list of care assets alongside prioritised black spots. These fed, in iterative cycles, directly into application prototyping: the most frequently reported categories became one-tap report types; care assets were turned into togglable layers; and critical routes informed the first hyper-cartographies and the subsequent validation with specialists. Participatory mapping thus served two functions at once — as a research instrument and as what one might call a prefigurative device for democratising common space. As method, it surfaced the micro-scale at which older adults' mobility and sociability are shaped, generating observable variables for spatial analysis. As a device, it created a scene of care in which situated expertise set the vocabulary and priorities, laying the groundwork for a digital infrastructure designed not only to measure but to convene co-governance of the city.

Digital module and system architecture

Responding to RQ2 — the question of how to translate situated knowledge into actionable inputs — the technological core of Ciudad Mayor took shape as a two-layer socio-technical ecosystem developed through iterative co-design. The first layer is a native Android mobile application (prototype released on Google Play) serving as the interface for situated data capture (Figure 1); the second is a web platform for cartographic analysis and visualisation built on OpenStreetMap (OSM), functioning as a back office for municipal teams, researchers and project administrators, with differentiated user profiles. Early in the process, the initial observatory concept was reframed as an analytics platform in response to privacy concerns: the system stores georeferenced reports in anonymised form under explicit terms of use. Source code and the full system were transferred to the Santiago Regional Government at Technology Readiness Level 6.

Rather than cataloguing the full technical architecture, the focus here is on what co-design actually produced. The app captures two coordinate sets — the user's position at the time of reporting and the marker's location — together with structured metadata (title, comment, latitude, longitude, address, municipality, date and time, object identifier, and category). These fields were standardised to allow keyword and municipality filtering and to enable CSV downloads from administrator and municipal profiles. OpenStreetMap was adopted as the basemap to avoid licensing fees and to make multi-municipality deployment feasible. Via login, the analytics platform offers real-time hyper-

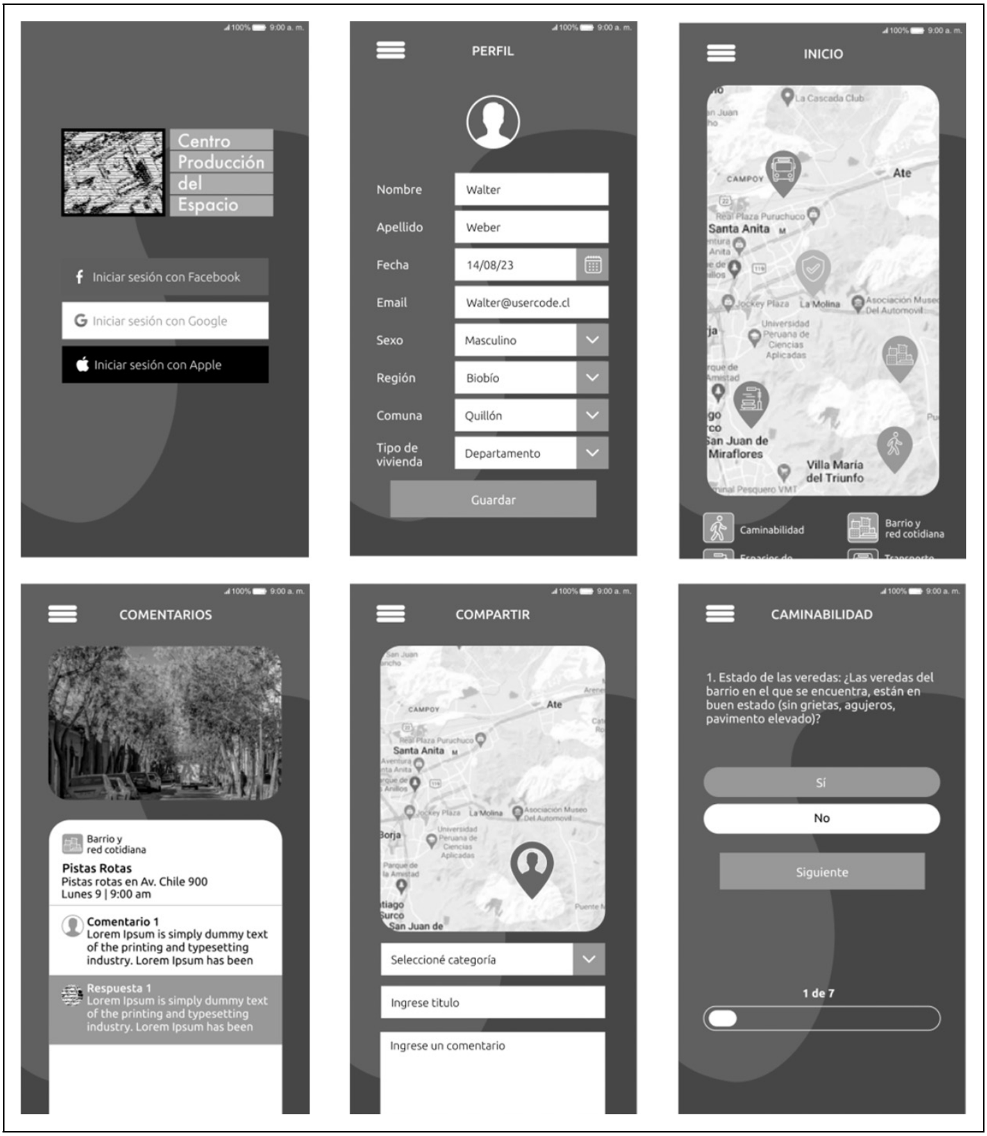

Figure 2. Real-time hyper-cartographies rendered on the web platform.
Source: Author.

cartographies (Figure 2), searchable and filterable tables, and data export (Figure 3), accompanied by user manuals and tutorials.

Three operational profiles were defined: Administrator/Research, with full repository access and download tools; Municipal, with territorial filters and query functions to support day-to-day management; and Older User, designed for minimal interaction — guided reporting, a simplified interface and

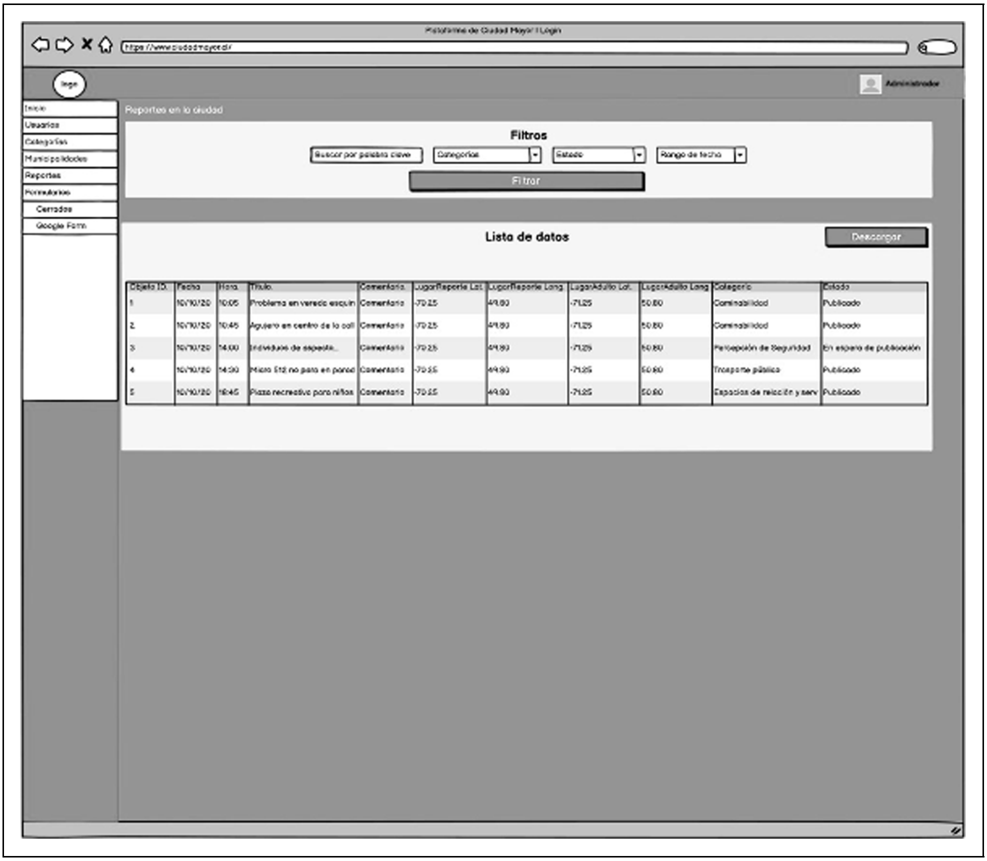

Figure 3. Tabular views enabling data export.
Source: Author.

visual aids — consistent with good practice for older populations and managed cognitive load.

Development followed an iterative co-design itinerary with communities and municipal teams, moving through: definition of principles and requirements; functional mock-ups; internal field trials using the administrator profile; and user-experience testing with older adults. Collectively, this process advanced the system from TRL-3 (laboratory prototype) to TRL-6 (prototype validated in a relevant environment). It entailed creating Apple/Android distribution accounts, configuring Secure Sockets Layer (SSL) certificates and secure-storage policies; Android was ultimately prioritised given its prevalence in the study communities and lower barriers to broader deployment.

The platform and app generated hyper-cartographies of perceived accessibility, safety and everyday infrastructure, layered over OSM and used in public deliberation settings — including hearings and meetings with local governments and the Regional Government during City Week. In parallel, user manuals were prepared and a public portal developed for access to the analytics platform. The system produces indicators sensitive to spatial variation (hyper-cartographies) and measurable via georeferenced reports, consistent with WHO criteria on measurability and replicability for age-friendly environments, while accommodating the context-dependence that the WHO itself underscores. The pivot towards an analytics platform and the role-based access design reflect a precautionary stance on personal-data protection and risk minimisation for older adults.

Development followed an iterative itinerary from TRL-3 to TRL-6. Usability testing with older adults (n = 20) yielded concrete refinements. Testing showed that 92% of participants could quickly identify the app's main functions, 83% navigated between sections with ease, and 75% found interactive controls adequately sized. However, 42% reported that the application did not run smoothly — a problem attributed unanimously not to the interface but to connectivity infrastructure. Connectivity lag emerged as a prominent barrier during field trials, which shifted attention from individual digital literacy to the materiality of telecommunications networks as a prerequisite for inclusion.

Hyper-cartographies and integration with spatial indicators

Still addressing RQ2, the iteration between workshop rounds and platform development made possible the construction of hyper-cartographies that bring together citizen reports and objective layers of the urban environment. In the pilot municipality of Providencia, the platform compiled 147 records organised under walkability, safety, services and transport, allowing density analysis, outlier detection and spatio-temporal patterning of older adults' urban experience. These inputs were cross-referenced with secondary data to produce reproducible analytical maps — points of interest and streets from OpenStreetMap, tree canopy and green areas, and health and wellbeing facilities — generating actionable multivariate accessibility overlays that operationalise the 15-minute city logic for Santiago.

During the Providencia institutional pilot, 147 records were compiled in total, of which 146 were valid field entries generated by the research team during route testing along the Miguel Claro–Tobalaba corridor. The distribution of these reports captures the primary concerns along this high-density pedestrian axis: 46.6% related to walkability (degraded pavements, uneven surfaces, fall hazards and discontinuities), 31.5% to proximity services and public space quality (disused street kiosks, inadequate street furniture, deteriorated plazas), 15.8% to safety (dangerous crossings, inadequate or absent pedestrian signage, excessively short signal phases), and 6.1% to transport infrastructure (metro access conditions, bus stop design, intermodal connectivity).

Clusters of negative reports — potholes, degraded pavements, unsafe crossings — appeared repeatedly along the edges of metropolitan corridors and in areas of reduced tree cover, features which prior evidence associates with reduced willingness to walk among older people as well as heightened risks of falls and fear of crime. Conversely, clusters of positive reports were closely tied to the proximity of pharmacies, health centres and social spaces, echoing quality-of-life research that foregrounds networks, activities and housing.

Methodologically, the integration of perceptions and objective layers was built into the tool's architecture: profiles with filters by date, time, title, location and verification status; tables enriched with coordinates, addresses and comments to trace routes and proximities; and full CSV downloads for subsequent analysis and third-party scrutiny.

The same data can be exported from administrator and research profiles to derive further indicators — densities per kilometre of pavement, incidents per 1000 older residents — and to cross-audit against municipal asset inventories for maintenance and lighting. Conceptually, these hyper-cartographies show how the produced city and its everyday rhythms are inscribed in the experience of ageing: micro-morphological frictions (kerbs, gradients, shade) can function as barriers or supports to agency, while functional proximity intersects with belonging and care. By articulating situated data with standardised layers, the device avoids the one-size-fits-all approach that WHO guidance and comparative indicator assessments have criticised, emphasising instead contextual adaptation and caution against collapsing multiple domains into single aggregated indices.

Institutional adoption and data governance for care

Addressing RQ3, the project examined the institutional conditions required to sustain this digitally mediated commons. The approach centred on designing a complete municipal use chain: differentiated profiles for citizens, municipal staff and administrators/researchers; input and verification workflows; and operating manuals. Minutes from co-design sessions with municipal teams across Renca, Independencia, and Quinta Normal document a process in which functional mock-ups of the administrator panel were presented, recommendations gathered, and key refinements carried out — normalising fields for municipal filtering, establishing the need for staff training and scoping how far the tool's reach would extend before broader roll-out. Keyword search, municipality filtering, standardised coordinate and address fields, and evidence export were among the functions refined as a result.

The governance strategy included preparing user manuals for citizen and administrator profiles, treated as a prerequisite for transfer and operation and built in from the prototype stage. Manuals were scheduled for revision following the participatory cycle to support training and system continuity. Particular attention was also given to expectation management and to indexing app functions against existing municipal budget lines, so that prioritisation and expenditure traceability remained legible to municipal staff.

Rather than a technocratic solution that outsources decision-making or closes the data loop, this governance model maintains public traceability and formalises the municipal verification circuit. Securing intellectual-property arrangements to transfer the codebase to the Santiago Regional Government, the integrated system demonstrates clear potential for operational use — offering a pragmatic scaffold for prioritising low-barrier micro-interventions on the basis of contextualised data.

Discussion

The evidence is clear and compels us to reframe the debate. How much longer will we view smart cities through the narrow lens of managerial efficiency and technocratic solutionism? The findings of this research demand that we shift the discussion towards spatial justice and an ethics of care. By bringing the reviewed literature into dialogue with our empirical results, three dialectical reflections emerge that challenge the current model of urban management.

Firstly, regarding micro-scale frictions (RQ1), the situated evidence — constructed from the ground up with older people — demonstrates that standardised indicators ultimately collude with institutional blindness. The literature had already warned of the dangers of generic, one-size-fits-all approaches (Rugel et al., 2022), but our results reveal how this lack of discernment manifests in everyday life: broken pavements, insufficient pedestrian crossings, or a lack of lighting that triggers a justified fear of gender-based violence. Reading this through a Lefebvrian lens, the finding is inescapable: when these embodied frictions are left out of official metrics, the indicator ceases to be a planning tool and becomes a mechanism of exclusion. Nothing could be further from reality than the promises of the '15-minute city' if it is not accompanied by a human-scale infrastructure of care; without this, the right to the city is merely a dead letter.

Secondly, in exploring the translation of experiential knowledge (RQ2), the results of the co-design process directly challenge those who blame older people for the digital divide. The dominant literature, particularly in Chile, tends to frame the problem around the need for individual 'literacy' (Riveros et al., 2020; Valenzuela Urra et al., 2022), assuming a kind of cognitive deficit on the part of the user. However, the high level of engagement with the platform demonstrated by participants showed that when technology is designed inclusively, citizen agency responds forcefully. The real frictions during the fieldwork arose from issues of latency and connectivity within neighbourhoods. This forces us to reorient the debate: digital exclusion in our cities is fundamentally a lack of material infrastructure. Without stable telecommunications networks guaranteed as a public good, urban digitalisation risks parcelling out benefits and deepening territorial inequality.

Turning to the institutional conditions required to sustain this model (RQ3), the traceability achieved at the municipal level offers a pragmatic pathway towards a genuine municipalism of care. Unlike hegemonic smart cities, which outsource decision-making and privatise the data cycle (Kolotouchkina et al., 2024), the architecture of Ciudad Mayor formalises a verification loop by keeping information under public scrutiny. By linking citizen reports directly to local funding streams, the tool acts as an institutional scaffold connecting informal solidarity networks (the 'shadow infrastructures' identified by Power et al., 2022) with formal governance. Stripped of its commercial guise, technology transcends the mere optimisation of flows to become a democratic interface for co-governance, capable of articulating territorial public policies with the embodied needs of the community.

Final reflections

This article shows that a care-oriented smart city is plausible when three analytical and operational layers are combined: situated knowledge production through participatory mapping with older adults; prototyping of a civic application to enable institutional feedback; and contextual geospatial metrics that relate micro-urban frictions to public decisions. Conceptually, this positions infrastructures of care as a bridge between the age-friendly agenda and smart-city debates, sidestepping uncritical indicator standardisation and technological solutionism. Methodologically, the study contributes a replicable procedure for weaving co-design with GIS, generating hyper-cartographies that surface morphological frictions — pavements, lighting, crossings and heat islands — that aggregated metrics tend to obscure. For public policy, two shifts appear salient. First, digital inclusion must be treated as material infrastructure — encompassing connectivity, human support and accessible design — rather than isolated individual training. This aligns with findings that unguided digitisation can widen divides and segment benefits. Second, participatory learnings must be articulated with regulatory and investment frameworks. This ensures that conventional urban strategies, even those aiming at public ends such as densification or affordability (López-Morales et al., 2025), do not generate adverse territorial externalities for older residents, nor unevenly distribute benefits across ageing populations with unstable residential trajectories. The Santiago case suggests several lines for comparative research and action. Future studies should assess the effects of digitally mediated care on mental health and social participation, carefully incorporating educational gradients and neighbourhood contexts. There is also a need to test the transferability of this protocol in other Latin American middle-income cities with differing land regimes, while broadening age-friendly metrics to include situated micro-environmental and digital-literacy dimensions. Ultimately, we propose that combining lived cartographies, civic-digital infrastructure and fine-grained adjustments to the walkable environment offers a meso-institutional route for aligning spatial justice with dignified ageing. Rather than merely optimising urban flows, this approach foregrounds a choreography of care — a smart city made democratic, bottom-up and situated, capable of translating embodied needs into arrangements of co-governance.

Acknowledgements

Many thanks to the municipal teams of Renca, Independencia, and Quinta Normal, the older adult participants who contributed their time and local knowledge, and the Government of Santiago for facilitation during the 'Semana de la Ciudad' activities. The author wishes to express their gratitude to colleagues Leslie Landaeta, Carlos Aguirre, Juan Correa, Felipe Ulloa, and Francisca Cancino for their invaluable contributions to the fieldwork. Deep appreciation is also extended to Sara Ortiz Escalante and Blanca Valdivia Gutiérrez from Col·lectiu Punt 6; their inspiring book, Urbanismo Feminista, was fundamental in broadening the conceptual scope and reach of this research project.

ORCID iD

Francisco Vergara-Perucich https://orcid.org/0000-0002-1930-4691

Ethical considerations

All participants gave informed consent. Personal data were anonymised prior to analysis; no directly identifying information is reported.

Funding

The author disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This work was supported by the Gobierno Regional Metropolitano de Santiago through the Fondo de Innovación para la Competitividad Regional (FIC-R), Project 'Ciudad Mayor'; and by ANID-Chile grants FONDECYT 1262258, FONDECYT 1241297, and FONIS SA23I0032.

Declaration of conflicting interests

The author declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data availability statement

De-identified data derived from the Ciudad Mayor platform (reports and hyper-cartography exports) and documentation (user manuals) are available upon reasonable request to the corresponding author, subject to municipal data-sharing agreements and participant privacy safeguards.

Note

1. The eight WHO age-friendly topics are outdoor spaces and buildings; transportation; housing; social participation; respect and social inclusion; civic participation and

employment; communication and information; and community support and health services (WHO, 2007).

References

Azócar González R (2023) Masculinidades queer y colitud: experiencias de varones chilenos en el contexto neoliberal. *Anthropologica* 40(49): 192–210. https://doi.org/10.18800/anthropologica.202202.009

Bian X, Chen R and Jiang H (2026) Housing affordability and rent control: The case of elderly renters. *Urban Studies* 63(2): 372–394. https://doi.org/10.1177/00420980251359191

Branny A, Møller MS, Korpilo S, et al. (2022) Smarter greener cities through a social-ecological-technological systems approach. *Current Opinion in Environmental Sustainability* 55: 101196. https://doi.org/10.1016/j.cosust.2022.101196

Buffel T and Phillipson C (2018) A manifesto for the age-friendly movement: Developing a new urban agenda. *Journal of Aging & Social Policy* 30(2): 173–192. https://doi.org/10.1080/08959420.2018.1430414

Cox L, González R and Le Foulon C (2024) The 2019 Chilean social upheaval: A descriptive approach. *Journal of Politics in Latin America* 16(1): 68–89. https://doi.org/10.1177/1866802X23120374

Davis J (2022) *The Caring City: Ethics of Urban Design*. Bristol University Press.

De Santis KK, Mergenthal L, Christianson L, et al. (2023) Digital technologies for health promotion and disease prevention in older people: Scoping review. *Journal of Medical Internet Research* 25: e47501. https://doi.org/10.2196/47501

Fenster T (2005) The right to the gendered city: Different formations of belonging in everyday life. *Journal of Gender Studies* 14(3): 217–231. https://doi.org/10.1080/09589230500264109

Flores SL, Molina EH, Osuna JB, et al. (2018) All with you: A new method for developing compassionate communities—experiences in Spain and Latin-America. *Annals of Palliative Medicine* 7(S2): S15–S31. https://doi.org/10.21037/apm.2018.03.02

Gayo M and Méndez M (2022) Fragmentación ideológica de la elite en Chile. *Tempo Social* 34(2): 93–116. https://doi.org/10.11606/0103-2070.ts.2022.191178

Gomez-Hernandez M, Ferre X, Moral C, et al. (2023) Design guidelines of mobile apps for older adults: Systematic review and thematic analysis. *JMIR mHealth and uHealth* 11: e42527. https://doi.org/10.2196/42527

Green G, Jackisch J and Zamaro G (2015) Healthy cities as catalysts for caring and supportive environments. *Health Promotion International* 30(Suppl 1): i99–i107. https://doi.org/10.1093/heapro/dav037

Herrmann-Lunecke MG, Figueroa-Martínez C, Parra Huerta F, et al. (2022) The disabling city: Older persons walking in central neighbourhoods of Santiago de Chile. *Sustainability* 14(17): 11085. https://doi.org/10.3390/su141711085

Kern L (2020) *Feminist City: Claiming Space in a Man-Made World*. Verso.

Kim K, Buckley T, Burnette D, et al. (2020) Measurement indicators of age-friendly communities: Findings from the AARP age-friendly community survey. *Innovation in Aging* 4(Suppl 1): 377. https://doi.org/10.1093/geroni/igaa057.1213

Kolotouchkina O, Ripoll González L and Belabas W (2024) Smart cities, digital inequalities, and the challenge of inclusion. *Smart Cities* 7(6): 1–20. https://doi.org/10.3390/smartcities7060130

Kussy A, Palomera D and Silver D (2022) The caring city? A critical reflection on Barcelona's municipal experiments in care and the commons. *Urban Studies* 60(11): 2036–2053. https://doi.org/10.1177/00420980221134191

Lefebvre H (1991) *The Production of Space*. Blackwell.

Lefebvre H (2004) *Rhythmanalysis: Space, Time and Everyday Life*. Continuum.

Li Y and Zhao D (2021) Education, neighbourhood context and depression of elderly Chinese. *Urban Studies* 58(16): 3354–3370. https://doi.org/10.1177/0042098021989948

Liu Y, Dijst M, Faber J, et al. (2017) Healthy urban living: Residential environment and health of older adults in Shanghai. *Health &*

Place 47: 80–89. https://doi.org/10.1016/j.healthplace.2017.07.007
López-Morales E, Caimanque R, Herrera N, et al. (2025) The politics of vertical densification in Chile: Bridging planning, contestation and housing welfare under progressive municipalism. *Urban Studies* Epub ahead of print 11 November 2025. DOI: https://doi.org/10.1177/00420980251377714
Loyola E (2019) Acceso a la justicia y brecha digital en los adultos mayores. Informe sintético sobre la cuestión en Chile. *Trayectorias Humanas y Trascontinentales* 1(5): 123–136. https://doi.org/10.25965/trahs.1374
Marston HR, Dikken J, Waights V, et al. (2026) An intergenerational perspective of measuring age-friendliness across UK communities using an international measure. *Cogent Gerontology* 5(1): 2649060. https://doi.org/10.1080/28324897.2026.2649060
Masías Benavides Román A and Chipana Fernández YMM (2021) Competencias digitales en adultos mayores y acceso a la justicia: Una revisión sistemática. *Revista de Derecho* 6(1): 182–194. https://doi.org/10.47712/rd.2021.v6i1.121
Monte-Mór RL (2005) What is the urban in the contemporary world? *Cadernos de Saúde Pública* 21(3): 942–948. https://doi.org/10.1590/S0102-311X2005000300030
Moreno C (2020) *Droit de Cité: De la 'ville-monde' à la 'ville du quart d'heure'*. Observatoire.
Osorio Parraguez P, Torrejón MJ and Anigstein MS (2011) Calidad de vida en personas mayores en Chile. *Revista MAD* (24): 61–75. https://doi.org/10.5354/0719-0527.2011.13531
Pelfini A, Riveros C and Aguilar O (2023) ¿Han aprendido la lección? Las élites empresariales y su reacción ante las reformas. Chile 2014-2020. *Izquierdas* 52(41): 4738-4758. https://doi.org/10.4067/s0718-50492023000100241
Phillipson C, Bernard M, Phillips J, et al. (1999) Older people's experiences of community life: Patterns of neighbouring in three urban areas. *The Sociological Review* 47(4): 715–743. https://doi.org/10.1111/1467-954X.00196
Pinheiro F, Diogo M, Paúl C, et al. (2018) Age-friendly cities performance assessment indicators system hierarchization. *Revista Kairós: Gerontologia* 21(4): 31–54. https://doi.org/10.23925/2176-901x.2018v21i4p31-54
Pojani D (2021) *Trophy Cities: Why a Feminist Perspective on New Capital Cities?* Edward Elgar. https://doi.org/10.4337/9781839100775
Power ER, Wiesel I, Mitchell E, et al. (2022) Shadow care infrastructures: Sustaining life in post-welfare cities. *Progress in Human Geography* 46(5): 1321–1343. https://doi.org/10.1177/03091325221086208
Ramírez-Pereira M, Pérez Abarca R and Machuca-Contreras F (2020) Políticas públicas de promoción de salud en el contexto de la COVID-19, en Chile, una aproximación desde el análisis situacional. *Global Health Promotion* 28(1): 127–36. https://doi.org/10.1177/1757975920978311
Ravensbergen L, Van Liefferinge M, Isabella J, et al. (2022) Accessibility by public transport for older adults: A systematic review. *Journal of Transport Geography* 103: 103392. https://doi.org/10.1016/j.jtrangeo.2022.103408
Riveros C, Arenas Á, Castro M, et al. (2020) Public policies, gaps and digital literacy of the elderly people. *Law, State and Telecommunications Review* 12(1): 137–158. https://doi.org/10.26512/lstr.v12i1.31180
Rojas V, Soto J, Cuadros V, et al. (2021) Vivencias y sentido de vida del adulto mayor víctima de violencia familiar en tiempos de COVID-19. *Revista Universidad y Sociedad* 13(4): 499–504.
Rugel EJ, Chow CK, Corsi DJ, et al. (2022) Developing indicators of age-friendly neighbourhood environments across 20 countries. *BMC Public Health* 22: 1263. https://doi.org/10.1186/s12889-022-13657-3
Sanz MF, Acha BV and García MF (2021) Co-design for people-centred care digital solutions: A literature review. *International Journal of Integrated Care* 21(2): 16. https://doi.org/10.5334/ijic.5573
Smith RJ, Lehning AJ and Kim K (2018) Aging in place in gentrifying neighborhoods: Implications for physical and mental health. *The Gerontologist* 58(1): 26–35. https://doi.org/10.1093/geront/gnx105
Ulloa-León F, Correa-Parra J, Vergara-Perucich F, et al. (2023) '15-Minute City' and elderly

people: Thinking about healthy cities. *Smart Cities* 6(2): 1043–1058. https://doi.org/10.3390/smartcities6020050

Valenzuela Urra C, Rodríguez Pastene F and Oliveros Castro S (2022) Gobernanza electrónica e inclusión digital de personas mayores mediante estrategias de alfabetización digital e informacional en la localidad de Placilla, Valparaíso, Chile. *Palabra Clave (La Plata)* 12(1): 1–26. https://doi.org/10.24215/18539912e168

van den Berg M (2012) Femininity as a city marketing strategy. *Urban Studies* 49(1): 153–168. https://doi.org/10.1177/0042098010396240

van Hoof J and Marston HR (2025) Who knows about the origins of the age-friendly cities and communities' topics? A critical analysis of the development process by the World Health Organization and a roadmap for an overhaul. *Cogent Gerontology* 4(1): 2602318. https://doi.org/10.1080/28324897.2025.2602318

van Hoof J, Kort HSM, Rutten PGS, et al. (2011) Ageing-in-place with the use of ambient intelligence technology: Perspectives of older users. *International Journal of Medical Informatics* 80(5): 310–331. https://doi.org/10.1016/j.ijmedinf.2011.02.010

van Hoof J, Marston HR, Kazak JK, et al. (2021) Ten questions concerning age-friendly cities and communities and the built environment. *Building and Environment* 199: 107922. https://doi.org/10.1016/j.buildenv.2021.107922

Vecchio G, Castillo B and Steiniger S (2020) Movilidad urbana y personas mayores en Santiago de Chile: El valor de integrar métodos de análisis. *Revista de Urbanismo* (43): 26–45. https://doi.org/10.5354/0717-5051.2020.57090

Von Humboldt S, Low G and Leal I (2022) Health service accessibility, mental health, and changes in behavior during the COVID-19 pandemic: A qualitative study of older adults. *International Journal of Environmental Research and Public Health* 19(7): 1–13. https://doi.org/10.3390/ijerph19074277

Woolrych R, Sixsmith J, Fisher J, et al. (2021) Constructing and negotiating social participation in old age: Experiences of older adults living in urban environments in the United Kingdom. *Ageing & Society* 41(6): 1398–1420. https://doi.org/10.1017/S0144686X19001569

Zhu Y and Diao M (2025) Effects of urban rail transit expansion on the mobility of the elderly: Findings from Singapore. *Urban Studies* 63(3): 457–483. https://doi.org/10.1177/00420980251359852